

Affordability Gap Updates

Week 6 - EdTrust Meeting

What we started with:

2,796 colleges and universities across the US

- Data merged from IPEDS, HUD, and BEA
- Financial variables included tuition, fees, and reported cost of attendance
- **Two variables dropped** due to 71% missingness
- More complete COA variables found on IPEDS to replace old
- **474 schools** missing all tuition data—**excluded** from gap calculations

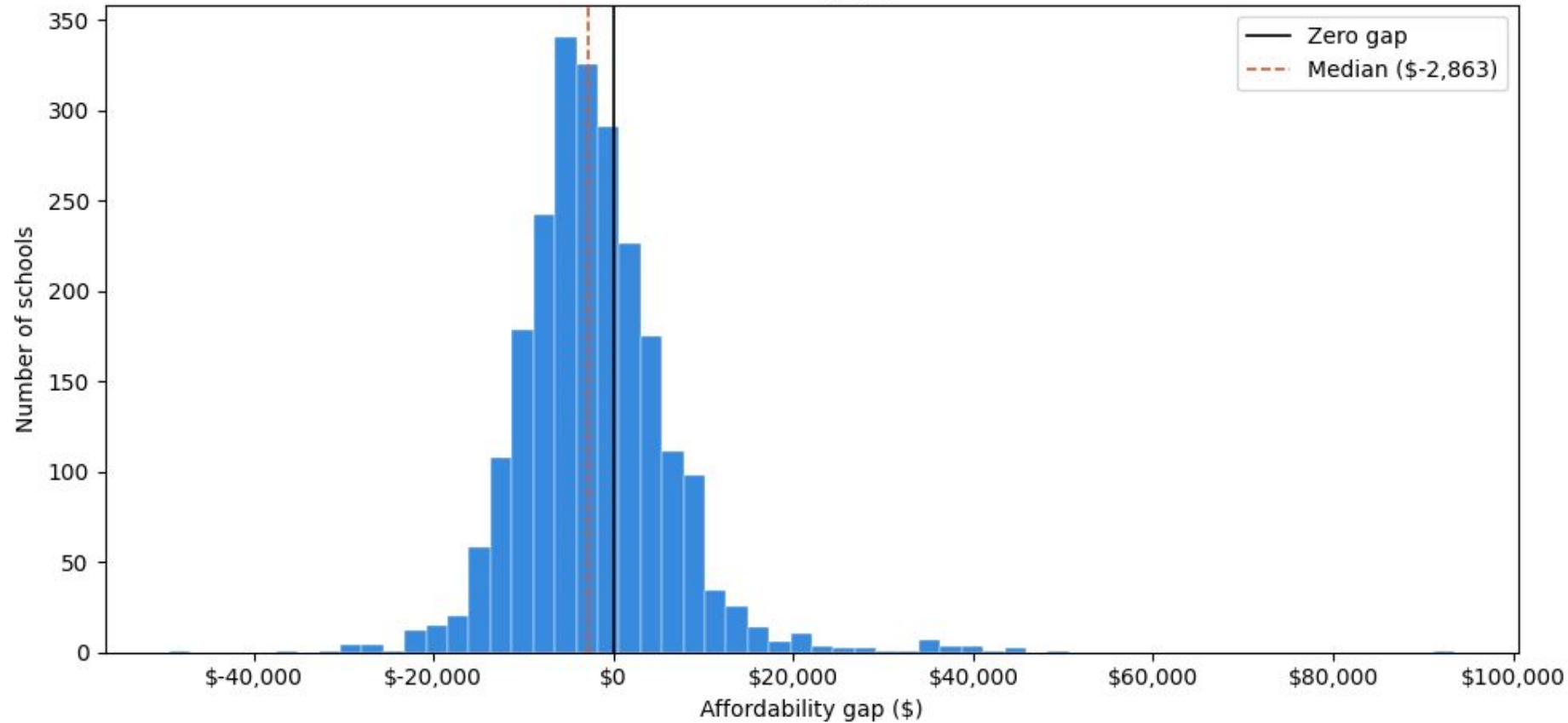
How we estimated actual costs

- **Food** — USDA Low-Cost Food Plan (March 2026), adjusted by state price level and locale type
- **Transportation** — BLS Consumer Expenditure Survey data, scaled by BEA region and locale type
- **Housing** — HUD Fair Market Rents, 1-bedroom, by school location
- All costs annualized on a 9-month academic year to match school reporting

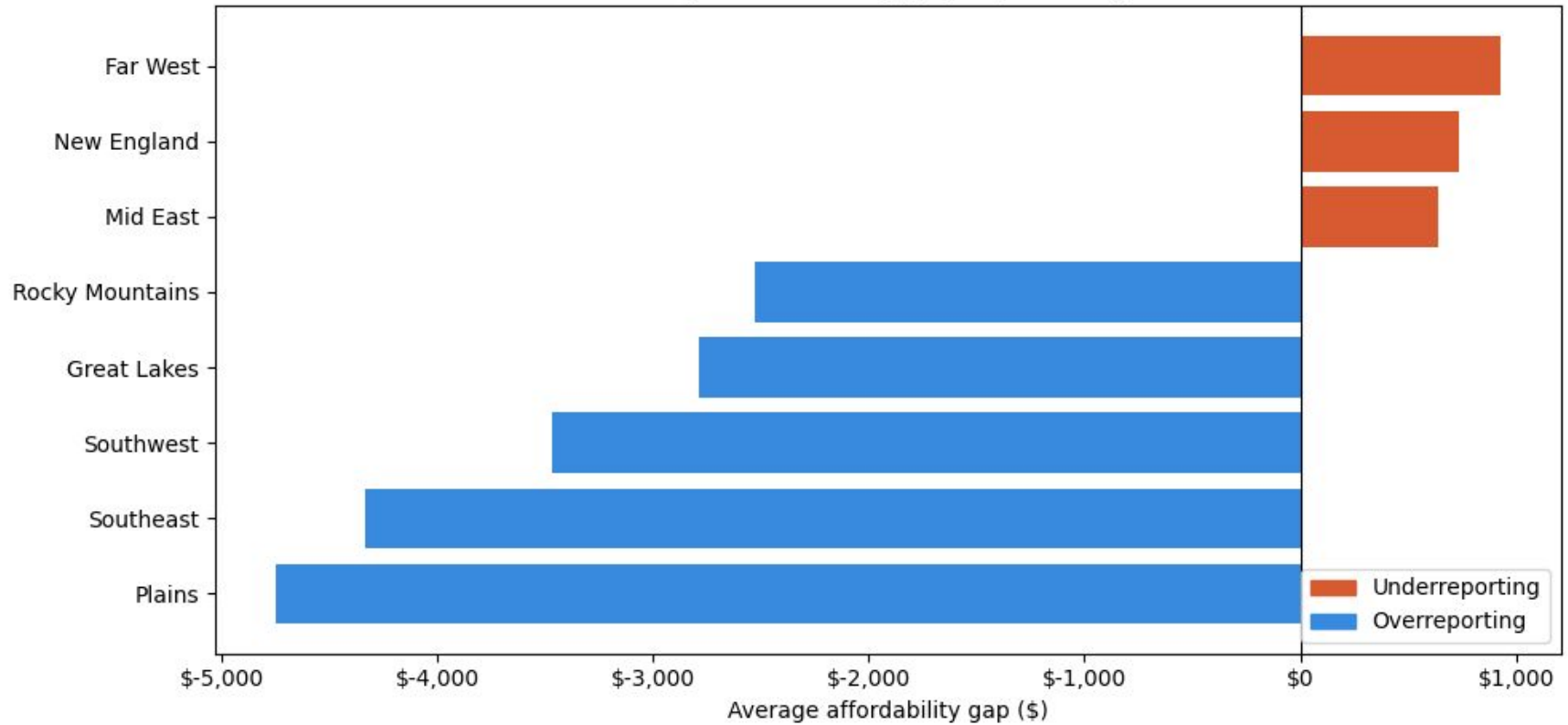
Key Data Decisions

- Used **9-month academic year** to match how schools report COA
- **Averaged male and female** USDA food figures across student population
- **Imputed missing rent** (106 schools) using median within same region and locale type — missingness was concentrated in small towns, not random
- Used **off-campus** room and board from IPEDS as the baseline for reported housing
- **Special focus schools** (theological, medical, law) **excluded** from gap analysis due to missing tuition data

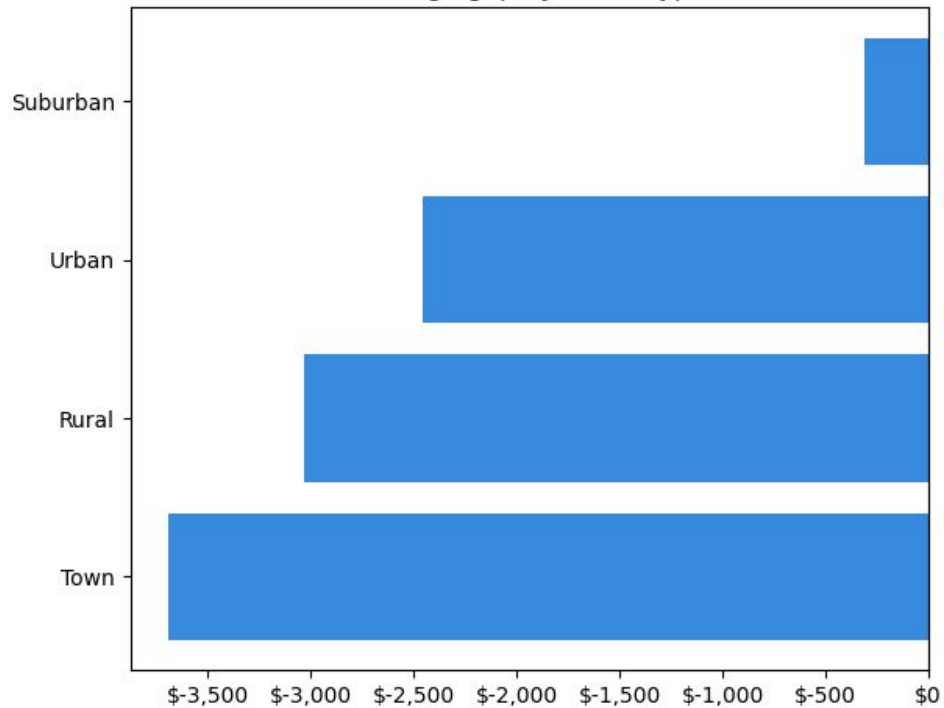
Distribution of affordability gap across all schools



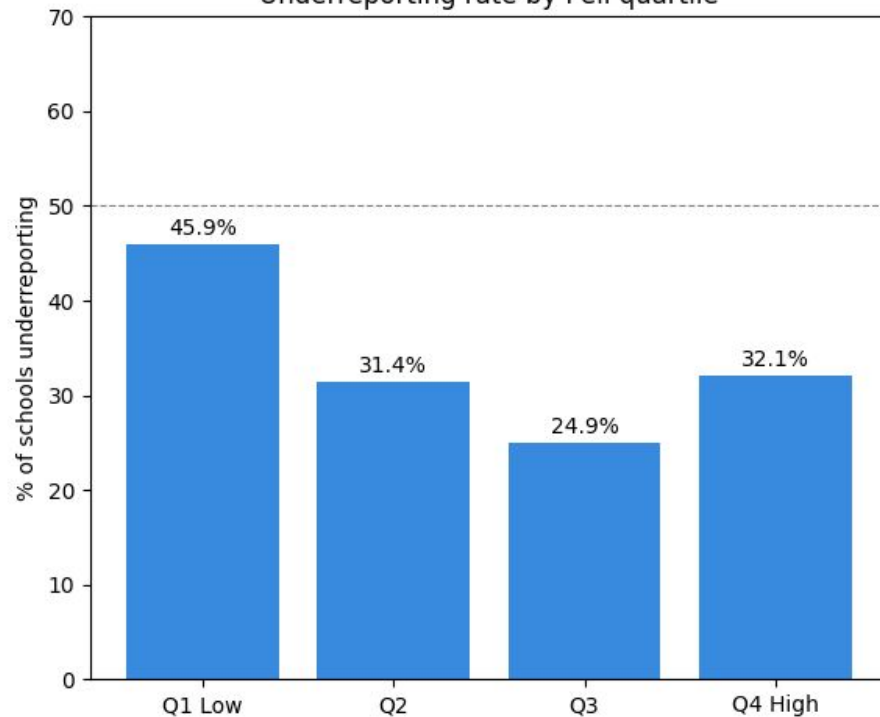
Average affordability gap by BEA region



Average gap by locale type



Underreporting rate by Pell quartile



Schools w/ Most Accurate Reporting

School	State	Locale	Reported COA	Est. Actual COA	Gap
University of Minnesota-Morris	Minnesota	Town	\$27,412	\$27,413	\$+1
Oglethorpe University	Georgia	Suburban	\$66,154	\$66,158	\$+4
SUNY at Purchase College	New York	Suburban	\$32,581	\$32,586	\$+5
College of the Holy Cross	Massachusetts	Urban	\$81,420	\$81,410	\$-10
Florida National University-Main Ca...	Florida	Suburban	\$39,065	\$39,046	\$-19
Drexel University	Pennsylvania	Urban	\$78,678	\$78,705	\$+27
Yeshiva Gedolah Keren Hatorah	New Jersey	Urban	\$33,900	\$33,928	\$+28
University of Miami	Florida	Suburban	\$83,000	\$82,966	\$-34
Art Academy of Cincinnati	Ohio	Urban	\$51,290	\$51,256	\$-34
Dallas Christian College	Texas	Suburban	\$39,285	\$39,249	\$-36

Modeling - true cost & over/underestimation

Baseline OLS - predicting 9mo affordability gap

- This model estimates which institutional characteristics are associated with larger or smaller affordability gaps.

- R-squared: 0.289, adjusted R-square 0.306
- 721 residuals

	coef	std err	z
Intercept	3443.9899	816.064	4.220
C(urban_group)[T.Rural]	-2204.8018	636.577	-3.464
C(urban_group)[T.Suburb]	734.7453	505.411	1.454
C(urban_group)[T.Town]	-2986.6533	413.733	-7.219
C(carnegie_group)[T.Baccalaureate]	-820.7289	623.710	-1.316
C(carnegie_group)[T.Doctorate]	-3704.7999	543.710	-6.814
C(carnegie_group)[T.Masters]	-1793.7244	571.318	-3.140
C(carnegie_group)[T.Other]	-1339.0258	1845.692	-0.725
C(carnegie_group)[T.Special Focus]	-1229.9756	736.096	-1.671
C(bea_region)[T.Great Lakes (IL, IN, MI, OH, WI)]	-2191.1588	641.631	-3.415
C(bea_region)[T.Mid East (DE, DC, MD, NJ, NY, PA)]	-2005.8474	830.556	-2.415
C(bea_region)[T.New England (CT, ME, MA, NH, RI, VT)]	-449.9049	936.876	-0.480
C(bea_region)[T.Other]	1.276e+04	1.03e+05	0.124
C(bea_region)[T.Other U.S. jurisdictions (AS, FM, GU, MH, MP, PR, PW, VI)]	-7968.3315	1468.970	-5.424
C(bea_region)[T.Plains (IA, KS, MN, MO, NE, ND, SD)]	-4172.9751	743.301	-5.614
C(bea_region)[T.Rocky Mountains (CO, ID, MT, UT, WY)]	-2345.4853	737.808	-3.179
C(bea_region)[T.Southeast (AL, AR, FL, GA, KY, LA, MS, NC, SC, TN, VA, WV)]	-4076.4982	635.474	-6.415
C(bea_region)[T.Southwest (AZ, NM, OK, TX)]	-3222.7241	642.582	-5.015
pct_pell	77.2688	13.119	5.890
Omnibus:	17.157	Durbin-Watson:	1.494
Prob(Omnibus):	0.000	Jarque-Bera (JB):	31.687
Skew:	0.102	Prob(JB):	1.32e-07
Kurtosis:	3.993	Cond. No.	1.34e+03

OLS (including rent) - predicting 9mo affordability gap

- This model estimates which institutional characteristics are associated with larger or smaller affordability gaps.

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- R-squared: 0.61, adjusted R-square 0.6

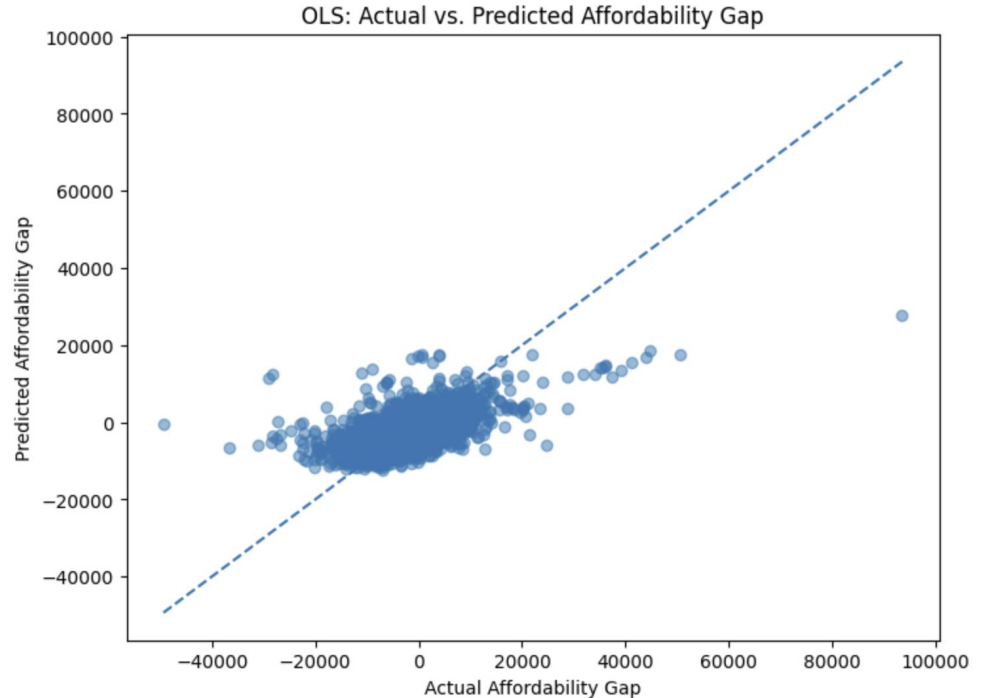
- 719 residuals

- 739 observations

	coef	std err	z
Intercept	-8.425e+04	3943.890	-21.363
C(urban_group)[T.Rural]	-940.8776	554.880	-1.696
C(urban_group)[T.Suburb]	-217.2831	366.028	-0.594
C(urban_group)[T.Town]	-851.2225	297.422	-2.862
C(carnegie_group)[T.Baccalaureate]	-30.0317	483.291	-0.062
C(carnegie_group)[T.Doctorate]	-3515.1002	427.190	-8.228
C(carnegie_group)[T.Masters]	-1553.0884	423.420	-3.668
C(carnegie_group)[T.Other]	2056.1715	1801.242	1.142
C(carnegie_group)[T.Special Focus]	-975.9728	545.625	-1.789
C(bea_region)[T.Great Lakes (IL, IN, MI, OH, WI)]	3775.2379	591.461	6.383
C(bea_region)[T.Mid East (DE, DC, MD, NJ, NY, PA)]	223.3661	596.973	0.374
C(bea_region)[T.New England (CT, ME, MA, NH, RI, VT)]	1385.7883	686.794	2.018
C(bea_region)[T.Other U.S. jurisdictions (AS, FM, GU, MH, MP, PR, PW, VI)]	3781.7028	1200.325	3.151
C(bea_region)[T.Plains (IA, KS, MN, MO, NE, ND, SD)]	1998.5483	715.022	2.795
C(bea_region)[T.Rocky Mountains (CO, ID, MT, UT, WY)]	826.3830	596.977	1.384
C(bea_region)[T.Southeast (AL, AR, FL, GA, KY, LA, MS, NC, SC, TN, VA, WV)]	362.4928	526.582	0.688
C(bea_region)[T.Southwest (AZ, NM, OK, TX)]	830.8900	511.528	1.624
C(bea_region)[T.U.S. Service schools]	1.004e+04	3.08e+04	0.326
pct_pell	62.4891	9.700	6.442
log_rent_1br	1.121e+04	523.581	21.415

OLS (updated CSV) - predicting 9mo affordability gap

- This model has 2,263 observations, making it more representative of the data we're looking for
- 0.365 r-squared, 0.351 adjusted
- Future: better to use, more representative of the actual data (even if it explains less variation).



Logistic regression

- This model predicts whether a school has a **larger** cost of attendance gap
- Includes Predictors for:
 - Urbanicity: rural, suburb, town, city
 - Carnegie classification
 - Region
 - Percent Pell Grant recipients
 - Shows the probability that a school underestimates living costs under the baseline affordability-gap measure

	coef	std err	z
Intercept	1.4349	0.431	3.330
C(urban_group)[T.Rural]	-0.4908	0.314	-1.564
C(urban_group)[T.Suburb]	0.4568	0.244	1.873
C(urban_group)[T.Town]	-1.0747	0.235	-4.575
C(carnegie_group)[T.Baccalaureate]	-0.5539	0.333	-1.663
C(carnegie_group)[T.Doctorate]	-1.9045	0.323	-5.898
C(carnegie_group)[T.Masters]	-1.2097	0.330	-3.664
C(carnegie_group)[T.Other]	-0.7308	1.179	-0.620
C(carnegie_group)[T.Special Focus]	-1.0366	0.382	-2.715
C(bea_region)[T.Great Lakes (IL, IN, MI, OH, WI)]	-0.3339	0.353	-0.945
C(bea_region)[T.Mid East (DE, DC, MD, NJ, NY, PA)]	-1.0637	0.351	-3.031
C(bea_region)[T.New England (CT, ME, MA, NH, RI, VT)]	0.1492	0.492	0.303
C(bea_region)[T.Other]	15.9661	5040.682	0.003
C(bea_region)[T.Other U.S. jurisdictions (AS, FM, GU, MH, MP, PR, PW, VI)]	-2.3530	0.641	-3.670
C(bea_region)[T.Plains (IA, KS, MN, MO, NE, ND, SD)]	-1.2516	0.401	-3.118
C(bea_region)[T.Rocky Mountains (CO, ID, MT, UT, WY)]	-0.9009	0.427	-2.108
C(bea_region)[T.Southeast (AL, AR, FL, GA, KY, LA, MS, NC, SC, TN, VA, WV)]	-1.2858	0.320	-4.024
C(bea_region)[T.Southwest (AZ, NM, OK, TX)]	-0.6498	0.368	-1.767
pct_pell	0.0229	0.006	4.058

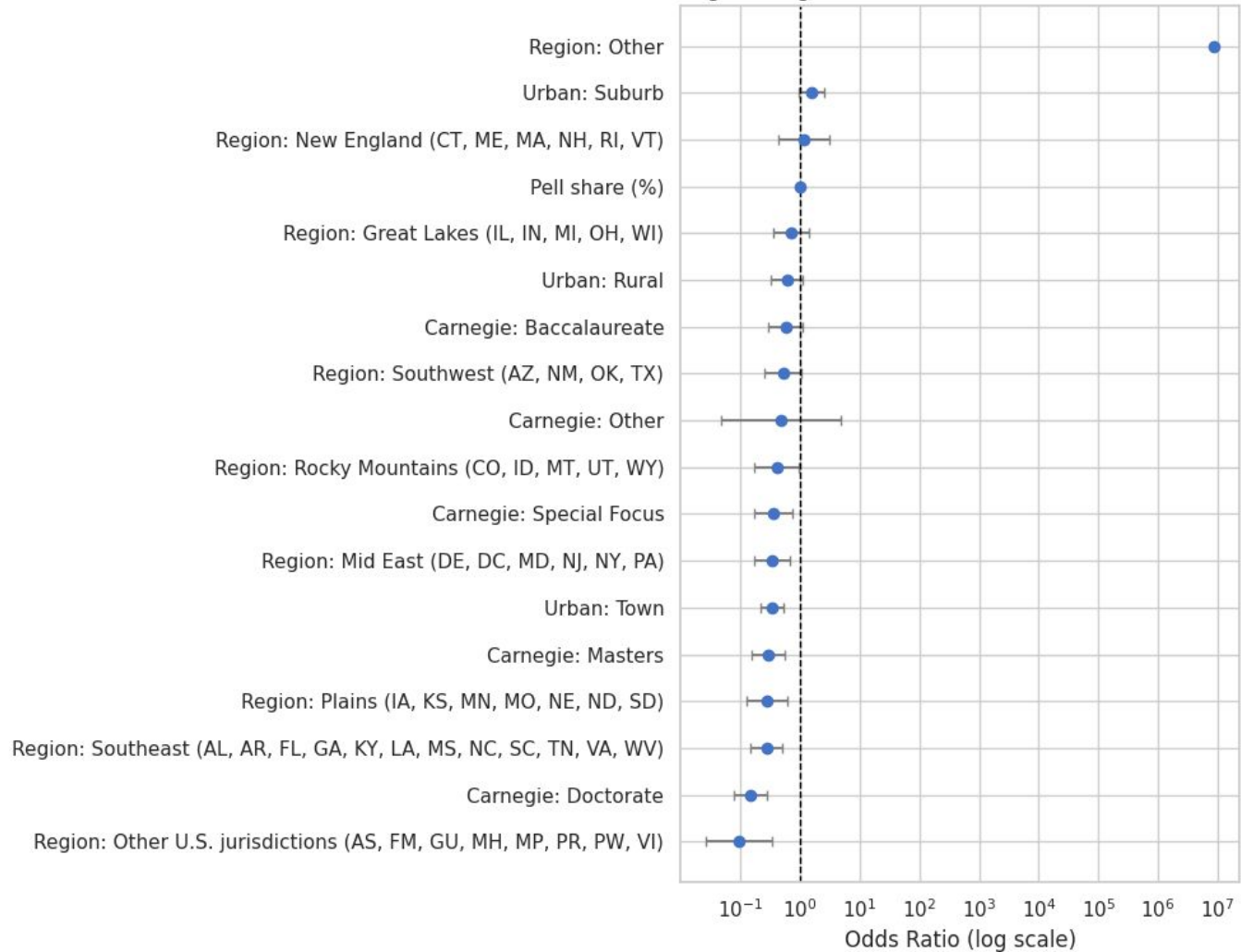
Log regression- does the school underestimate living costs

- 721 residuals
- 740 observations

Predictors with positive and significant Z-Scores

	coef	std err	z
Intercept	1.4349	0.431	3.330
C(urban_group)[T.Rural]	-0.4908	0.314	-1.564
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pct_pell	0.0229	0.006	4.058

Logistic Regression Odds Ratios for Underestimation



Descriptive summaries

	count	mean	std
reported_living_cost	803.0	13880.744707	4016.899028
actual_housing_cost_9mo	2690.0	12124.858610	4761.166700
food_cost_9mo_baseline	2796.0	3240.000000	0.000000
transport_cost_9mo_baseline	2796.0	1800.000000	0.000000
modeled_total_living_cost_9mo_baseline	2690.0	17164.858610	4761.166700
modeled_total_gap_9mo_baseline	740.0	2055.800138	5316.785112
modeled_total_gap_9mo_low	740.0	534.800138	5316.785112
modeled_total_gap_9mo_high	740.0	4206.800138	5316.785112
pct_pell	2321.0	45.722533	22.817863
rent_1br	2690.0	1347.206512	529.018522

Timeline for remaining project

Goal: create an interactive cost of attendance map that anyone can use on the Edtrust website (similar to the MIT living wage calculator)

Future steps: We want to use the modeling to perfect our dataset (example: if dropping rows with null values or adding a food and transportation dataset would improve the predictability/ accuracy of the dataset or not)

- Decision tree: interpretable nonlinear rules
- Random forest : feature importance + stronger prediction